

# Portfolio Daemon — Integration Points

**Project:** Portfolio Daemon

**Technology:** Python, asyncio, asyncssh, httpx, watchdog

**Location:** /home/dominic/Documents/portfolio-daemon

**Report #:** 17/20

| Integration     | Type          | Risk           |
|-----------------|---------------|----------------|
| SSH             | Remote access | Key management |
| Local processes | Service mgmt  | Port conflicts |
| File system     | Watching      | Permissions    |

## Integration Points

### Internal Integrations

| Source            | Target     | Protocol  | Data        | Frequency |
|-------------------|------------|-----------|-------------|-----------|
| Auth → Database   | SQLAlchemy | User data | Per request |           |
| Chat → OpenAI API | HTTP       | Messages  | Per request |           |
| Queue → WebSocket | WS         | Progress  | Real-time   |           |
| Frontend → API    | HTTP/WS    | All       | Per action  |           |

### External Integrations

| OpenAI | api.openai.com | HTTP/REST | AI chat completions | | GitHub Pages | github.io | HTTPS | Static site hosting | | Render | onrender.com | HTTPS | Backend hosting |

### API Integration Details

#### OpenAI API (AI Chat Proxy)

- **Endpoint:** POST `https://api.openai.com/v1/chat/completions`
- **Auth:** Bearer token (server-side key)
- **Protocol:** REST + SSE streaming
- **Rate limit:** 60 req/min (proxy level)
- **Models supported:** gpt-4, gpt-3.5-turbo
- **Timeout:** 60 seconds
- **Error handling:** 502 Bad Gateway on failure

## GitHub Pages

- **Service:** Static hosting
- **Protocol:** HTTPS + CDN
- **Auto-deploy:** Git push to master triggers rebuild
- **Used by:** WebBartosz, Reports

## Integration Health Monitoring

- All services expose `/health` endpoint
- WebBartosz polls every 60 seconds
- Render dashboards show service uptime
- No external monitoring service (e.g., Datadog, Grafana)

## Integration Testing

| Integration | Test Coverage | Notes                         |
|-------------|---------------|-------------------------------|
| Database    |               | Test with SQLite in-memory    |
| OpenAI      |               | Mocked in CI, live in prod    |
| WebSocket   |               | Tested manually               |
| Auth JWT    |               | Unit tests for token creation |